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Mechanical properties of expanded polystyrene lightweight concrete and brick

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ABSTRACT

In this study, mix proportion parameters of expanded polystyrene lightweight concrete are analyzed by using Taguchi's approach. The density, compressive strength and thermal conductivity were tested. The optimal mixture of EPS lightweight aggregate concrete was determined under consideration to manufacture the lightweight hollow brick. The most significant effect on compressive strength of EPS lightweight concrete is the cement ratio, while the content of cement and sand ratio has little effect. The relationship between density and compressive strength of EPS lightweight aggregate concrete is proposed as $f_c = 2.43 \times \gamma^{2.997} \times 10^{-9}$. The legitimacy of this relationship is confirmed.

1. Introduction

Recently, with the rapid development of high rise buildings, floating marine platforms, larger-sized and long-span concrete structures, lightweight concrete (LWC) has been a promising modern construction material [1–4]. In comparison with ordinary concrete, LWC shows some excellent characteristics such as lower density, higher specific strength, better thermal insulation and greater energy absorption which can be obtained by replacing standard aggregate totally or partially by lightweight aggregate (LWA).

Lightweight aggregate are broadly classified into two types: natural (pumice, diatomite, volcanic cinders, etc.) and artificial (perlite, clay, sintered fly ash, expanded shale, etc.). Expanded polystyrene (EPS) is a type of artificial lightweight aggregate with the density of only 10–30 kg/m³ [5]. Due to the ultra-lightweight density of EPS, the lightweight aggregate concrete is prone to segregation during casting. Many researchers have studied to overcome this drawback. Babu and Babu [6] added ultra fine silica fume to improve the bonding between EPS beads and cement paste. Roy et al. [7] utilized superplasticizer to avoid the segregation of EPS spheres. Chen and Liu [8] applied styrene-butadiene rubber (SBR) latex to improve the strength of concrete. Here, few special investigations of the EPS

can also provide an alternative development of EPS lightweight concrete considering the economy.

In this paper, the mechanical properties and thermal conductivity of EPS lightweight concrete are investigated. The proper proportion and the proper method which was a stationary, compressive strength and thermal conductivity are discussed. Then, the mechanical properties of EPS lightweight aggregate concrete and EPS lightweight bricks based on the cost, and thermal conductivity under compressive and

2. Experimental study

2.1. Materials

The cement produced by China is similar to ASTM type I. The maximum particle size is 0.075 mm and a fineness modulus of 2.2 and 10 mm. The commercially available lightweight aggregate is smooth and rounded shape which could be classified to specification GB/T 17431-2011